

Background

Microplastics are widely defined as fragments of plastic that are less than five millimeters (< 5mm) in diameter (Li et al., 2025).

Microplastics have been shown to have **adverse effects on the health of plants** (Fan et al.), **marine animals** (Huang et al.), and **humans** (Lin et al.).

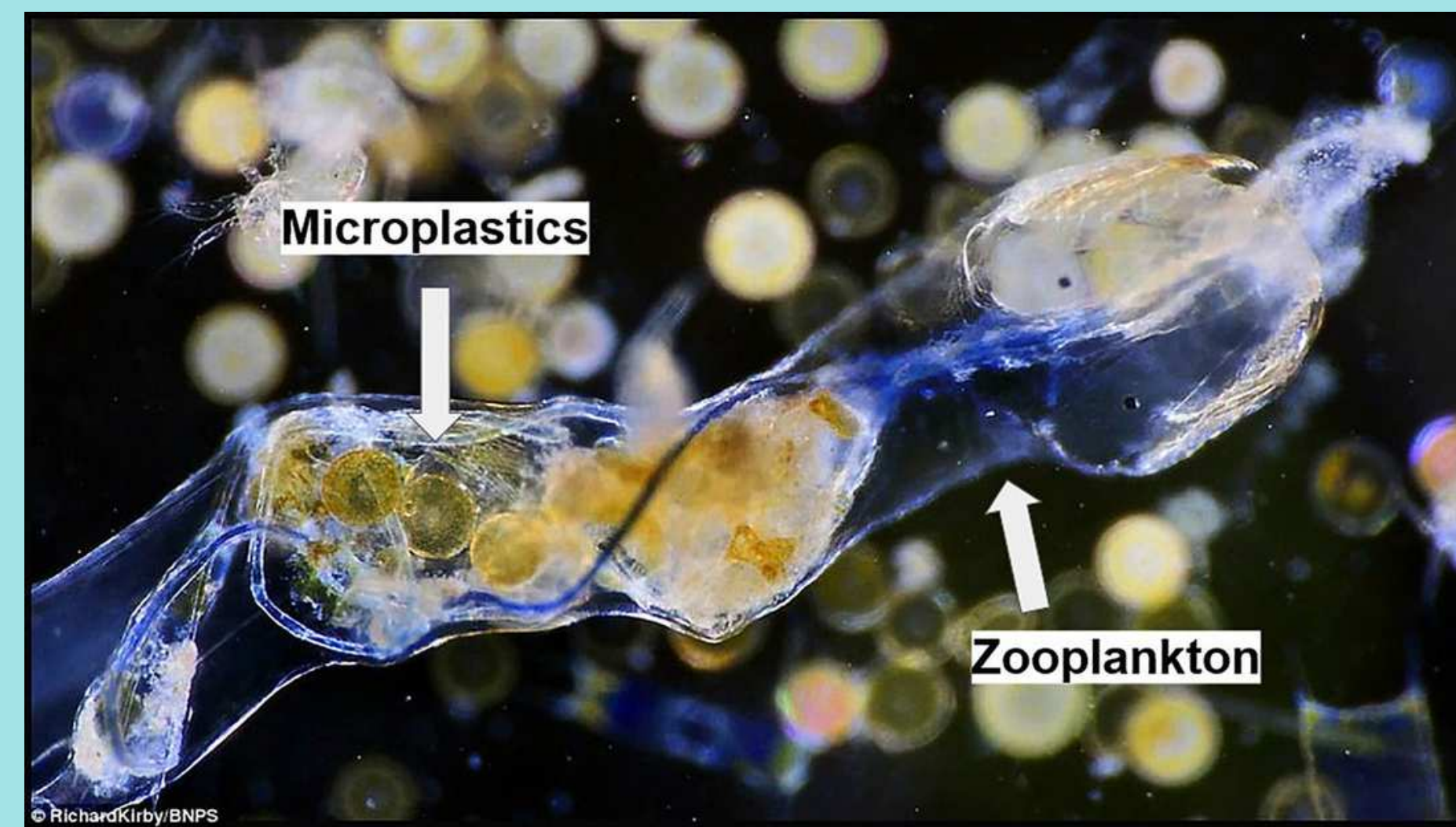


Figure 1: Microplastics inside a zooplankton (“The Nitty Gritty on Microplastics”)

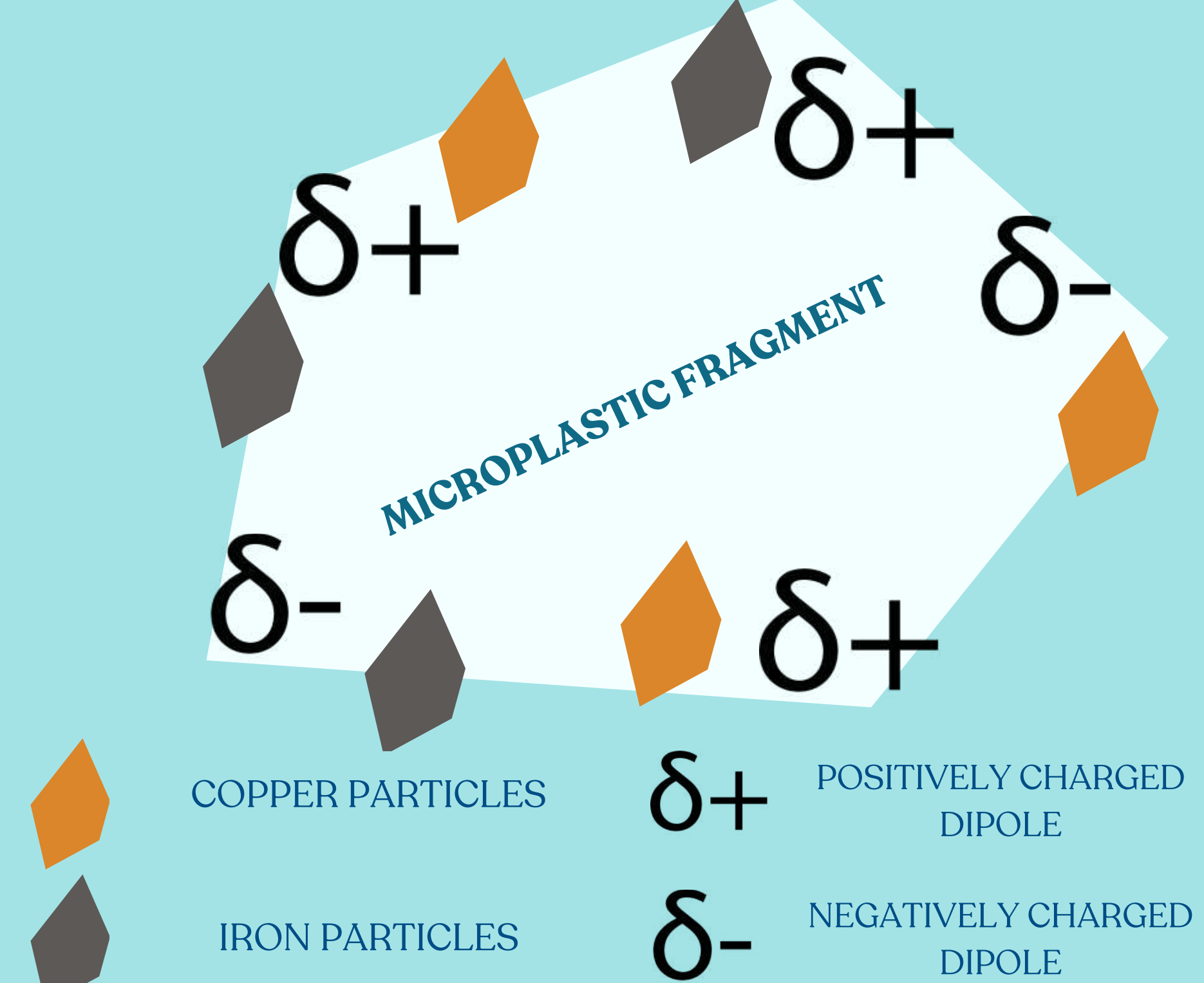


Figure 2: The adsorption of heavy metals onto the surface of microplastics (Created by researcher using Canva)

Due to their polar nature, heavy metals have been found to adsorb, or “stick”, onto the surface of microplastics (Hao et al., 2025).

Purpose

Using the polar nature of microplastics could allow us to **develop adsorption methods to remove them from water**, optimizing filtration. This could help remove microplastics from the food chain and **mitigate adverse effects**. This investigation utilizes hematite (Fe_2O_3) integrated into electrospun fibers in order to determine if an increase in hematite will increase polyethylene terephthalate (PET) microplastic filtration efficiency.

Cleaning our Water: Filtering Aqueous PET Microplastics Using Hematite- Based Electrospun Fibers

Methodology

1: Fabrication of microplastics

After being ground to dust in a spice grinder, PET plastic pellets were filtered through 425 nm and 212 nm sieves; the microplastics within this range were collected.

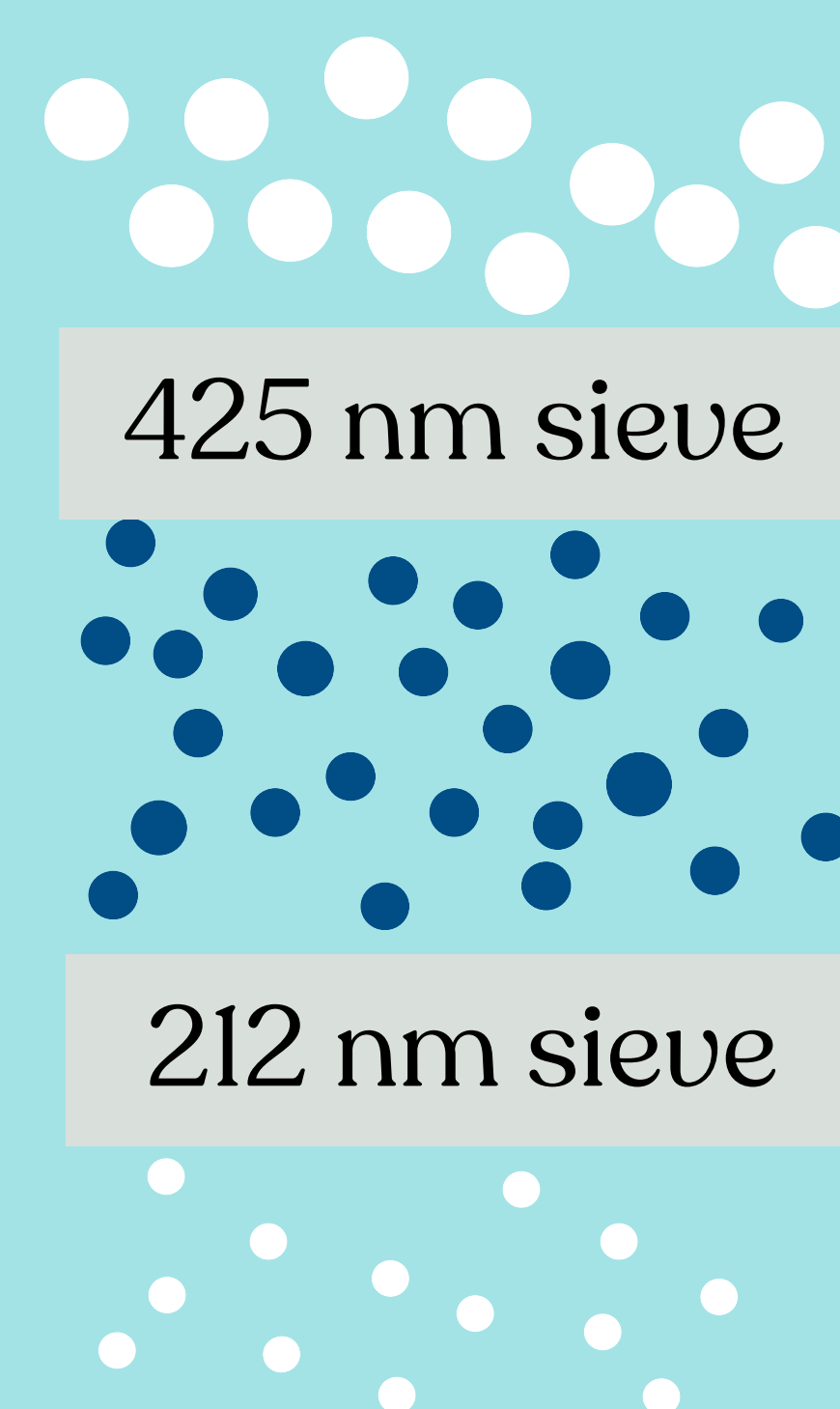


Figure 3: Production of PET Microplastics (Created by researcher using Canva)

3: Electrospinning

2 mL of the solutions were put into the electrospinning apparatus with a 23 gauge needle and electrospun onto an aluminum plate at 10 kV and 1 mL per hour.

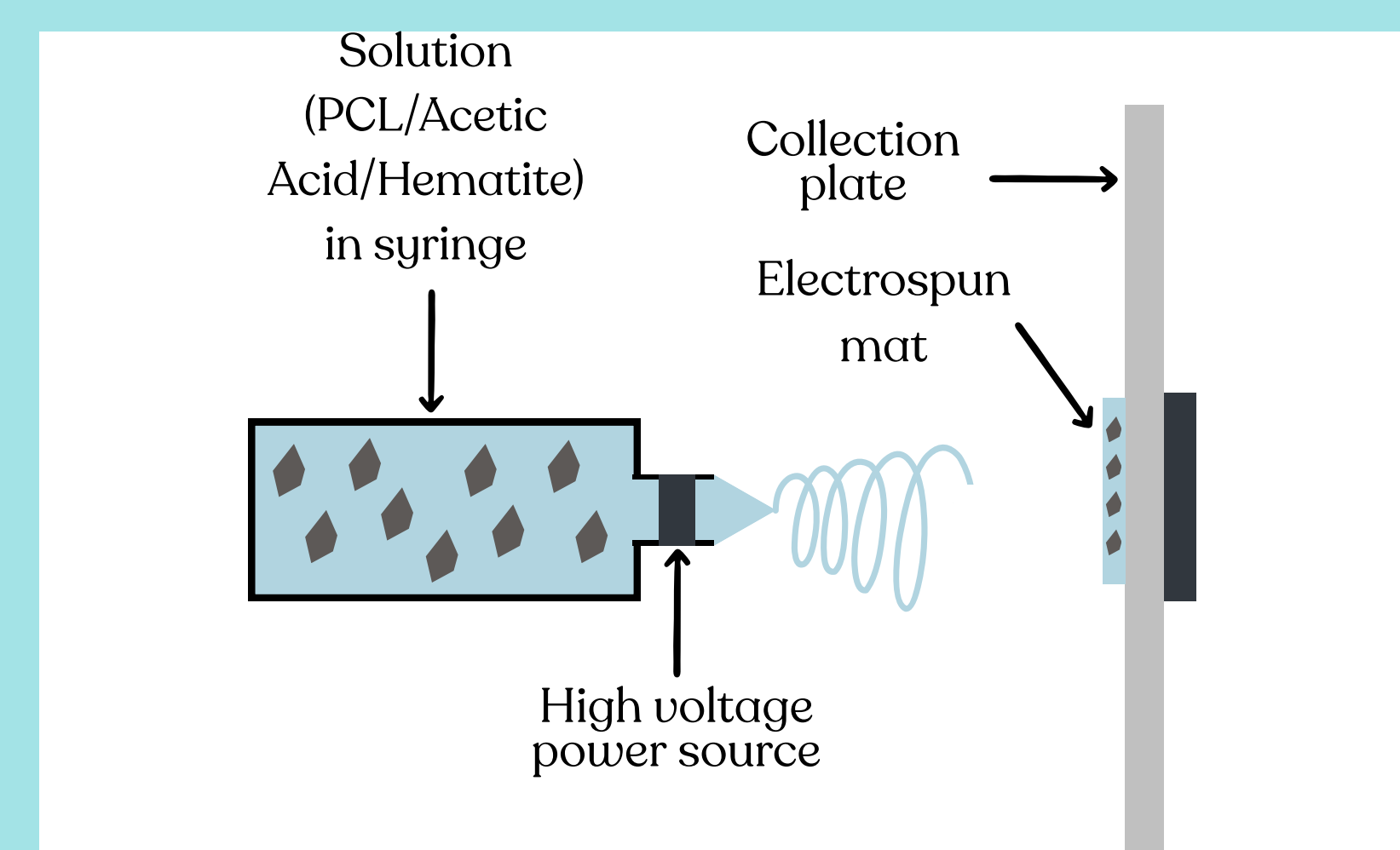


Figure 5: Electrospinning plates (Created by researcher using Canva)

2: Making the

electrospinning solution

0.75g of Polycaprolactone (PCL) pellets, 5mL of acetic acid, and masses of 0g (control), 0.02g, 0.04g, 0.06g, 0.08g, and 0.1g of hematite were dissolved together with a magnetic stirrer, creating a solution for electrospinning.

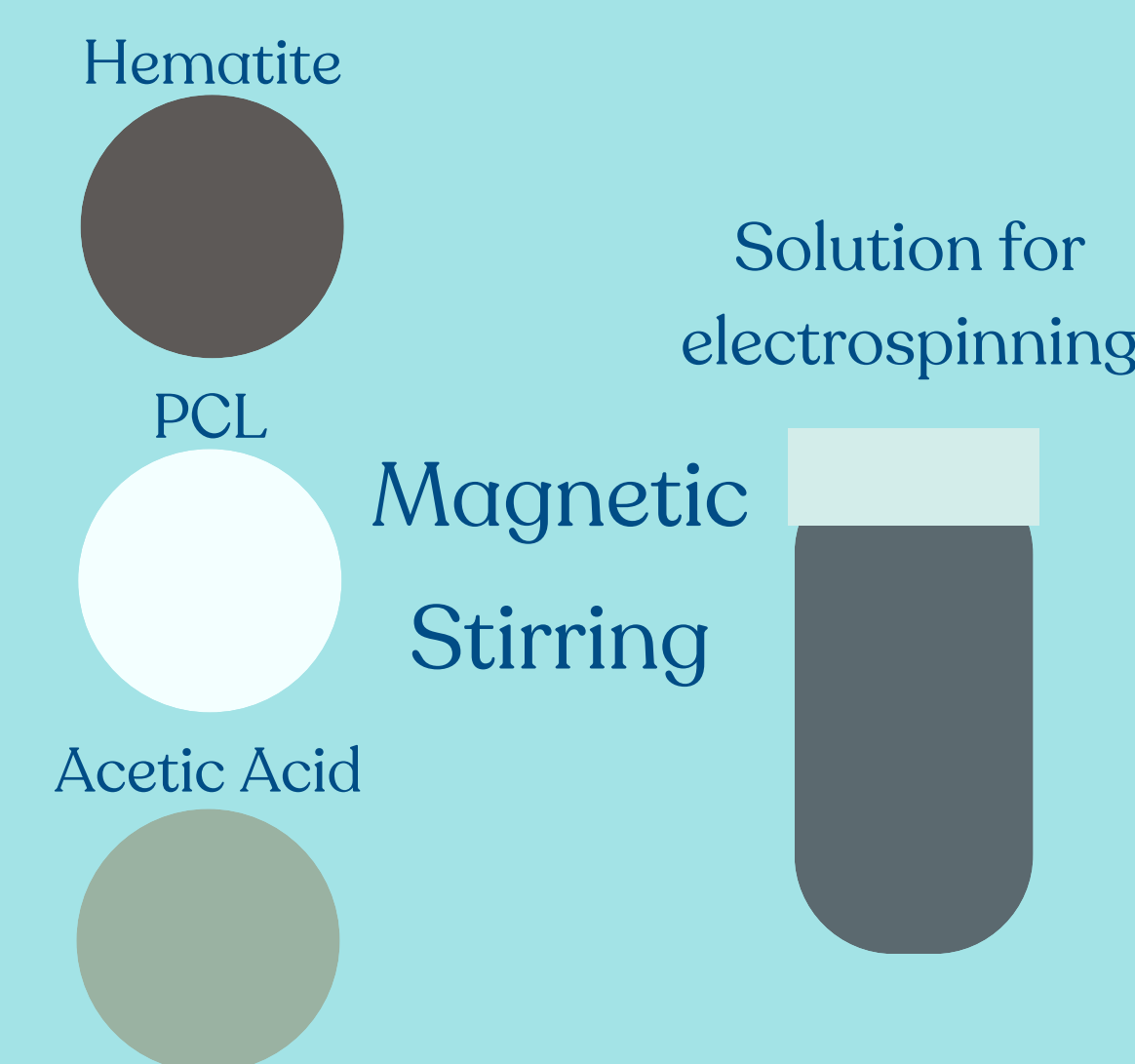


Figure 4: Creation of Hematite-Based Electrospinning Solution (Created by researcher using Canva)

4: Exposure to aqueous PET microplastics

1 g of the PET microplastics were placed into a beaker containing distilled water. 5cm x 5cm plates were exposed to the solution for 5 minutes, and the final weight of the microplastics was recorded.

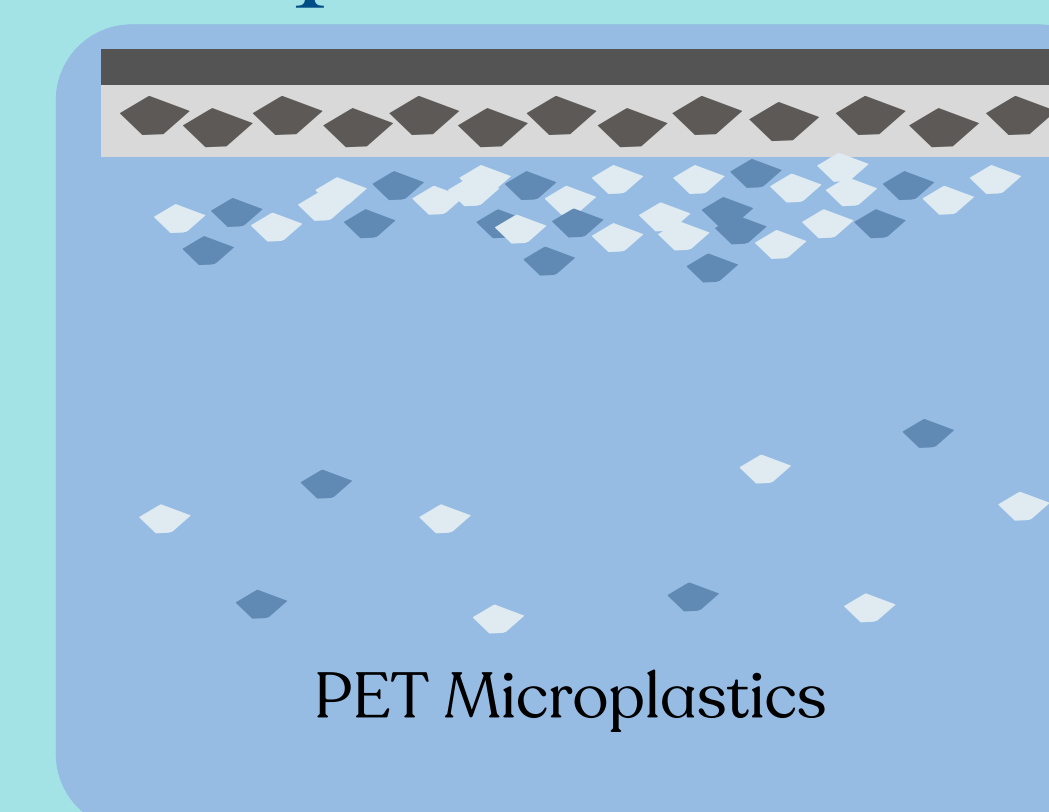
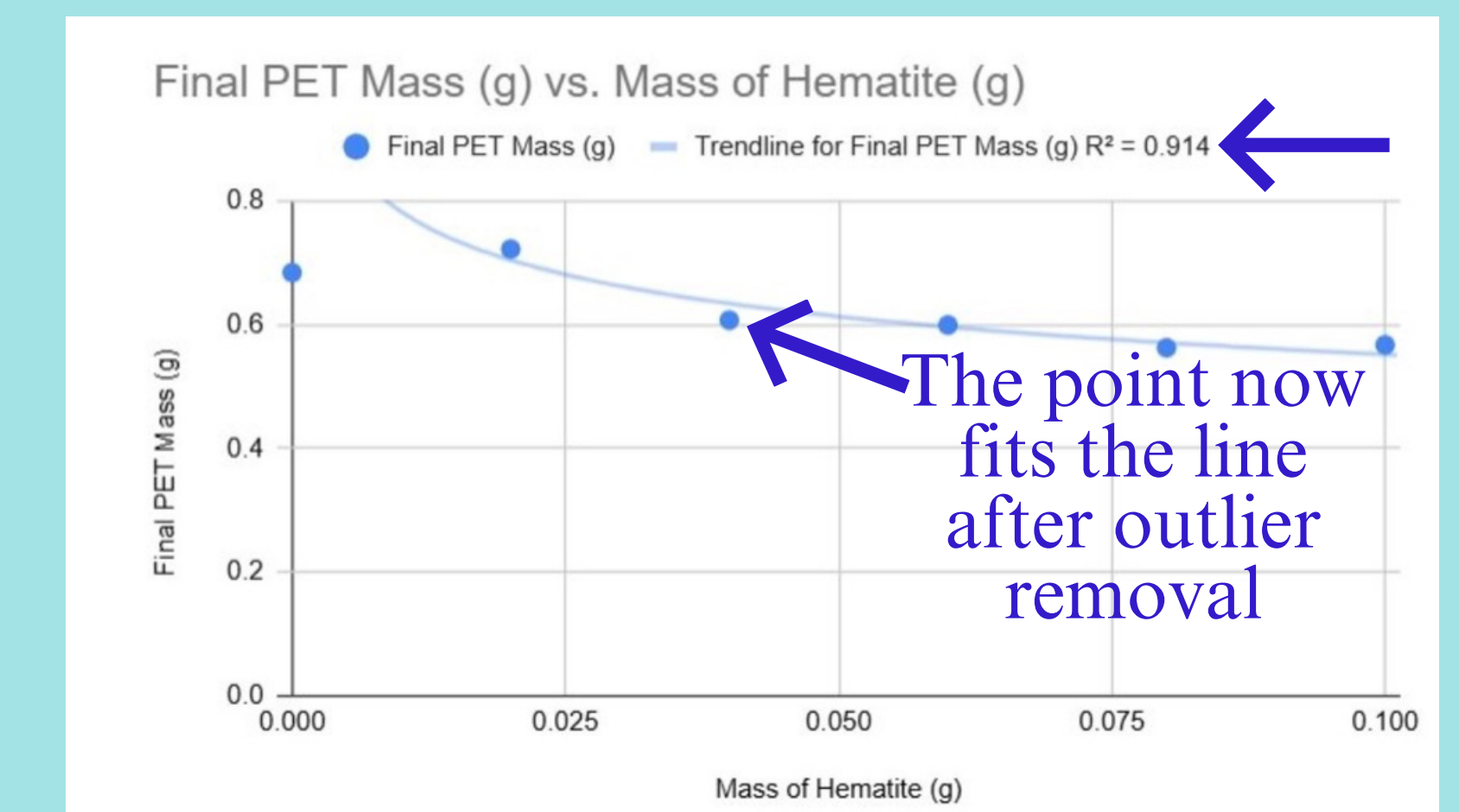
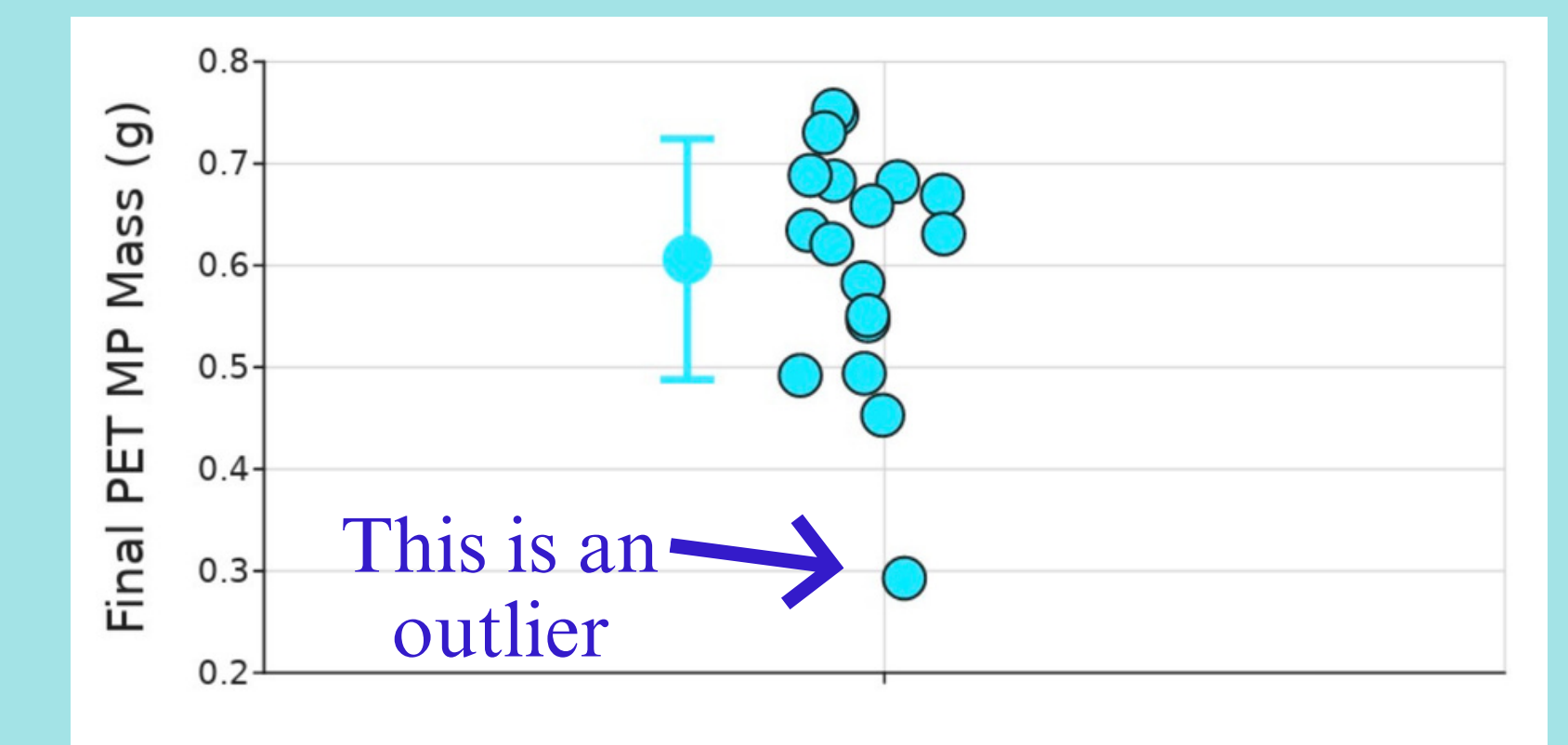
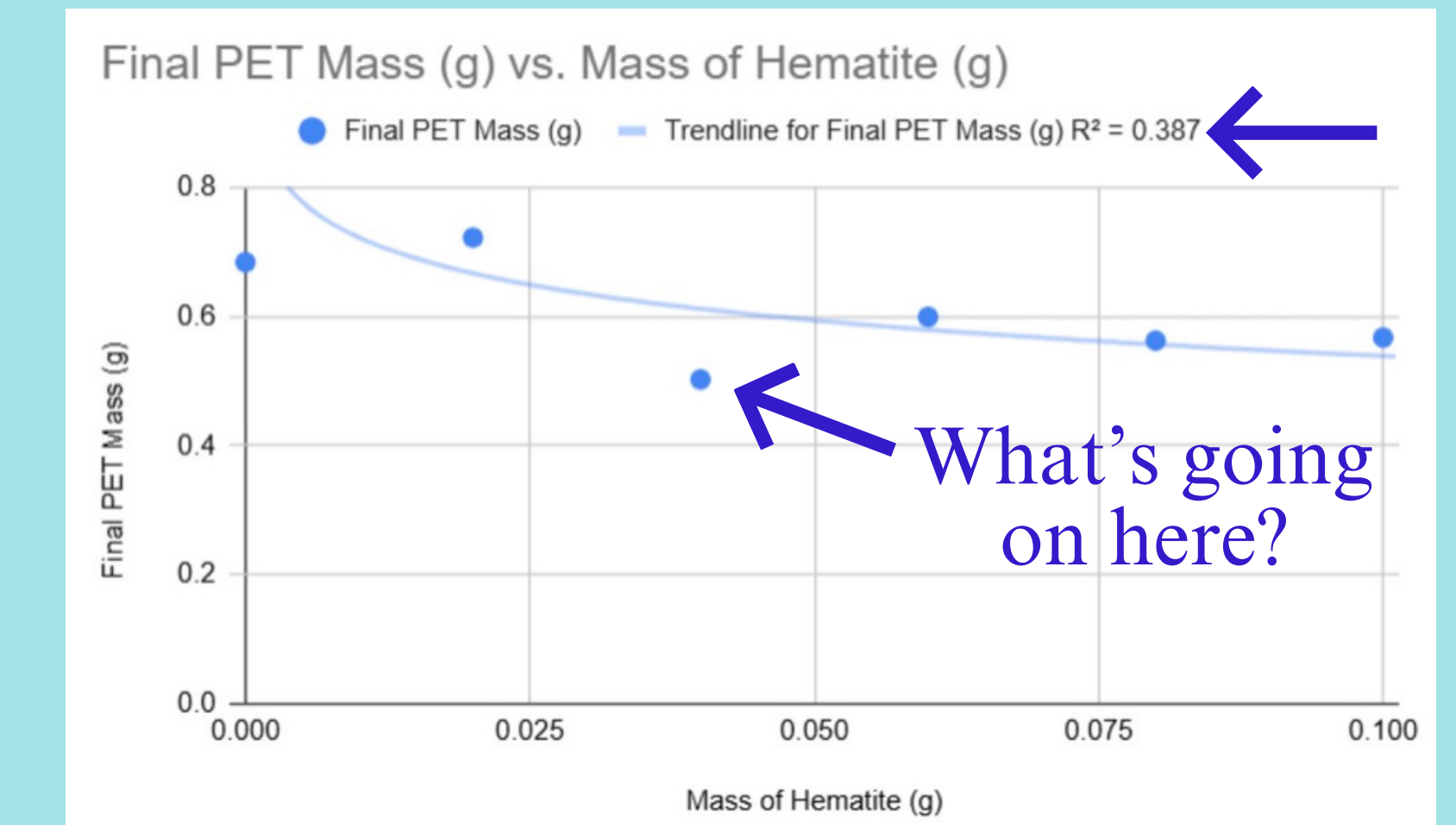


Figure 6: Plate exposure to microplastics (Created by Researcher using Canva)

Results



Conclusions

The final PET microplastic mass follows a **logarithmic decline** when the outlying data point is removed. This data point, existing for an 0.04g plate trial, was shown to be outlying based on a Z-Score statistical test, deviating almost three standard deviations from the mean. However, based on a Kruskal-Wallis statistical test, the **final masses of PET microplastics were not shown to have any statistically significant differences** between themselves.

Future Extensions

In the future, possible independent variables include different types of heavy metals (copper, other variations of iron, etc), different variations of plastics, adjusting electrospinning settings, and continuing the increase hematite concentrations

References

- Ashrafy, A., Liza, A. A., Islam, M. N., Billah, M. M., Arafat, S. T., Rahman, M. M., & Rahman, Sk. M. (2022). Microplastics Pollution: A Brief Review of Its Source and Abundance in Different Aquatic Ecosystems. *Journal of Hazardous Materials Advances*, 9, 100215. <https://doi.org/10.1016/j.hazadv.2022.100215>
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- Brennecke, D., Duarte, B., Paiva, F., Caçador, I., & Canning-Clode, J. (2016). Microplastics as vector for heavy metal contamination from the marine environment. *Estuarine, Coastal and Shelf Science*, 178, 189–195. <https://doi.org/10.1016/j.ecss.2015.12.003>